

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

2BI2__ / 2IPBI2__

REGISTRATION
NUMBER

BIOLOGY

9648/02

Paper 2

1 September 2014

2 hours

Additional Materials: Answer Paper and Cover Page

READ THESE INSTRUCTIONS FIRST

Write your Biology class, registration number and name on all the work you submit.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

The number of marks is given in brackets [] at the end of each question or part question.

Section A to C

Answer all questions.

Write your answers in this booklet.

Section D

Write your answers on the separate Answer Paper.

Begin each section of the question on a new piece of paper.

For Examiner's Use	
Section A - C	[Total: 80]
1	/ 11
2	/ 8
3	/ 12
4	/ 11
5	/ 11
6	/ 6
7	/ 11
8	/ 10
Section D	[Total: 20]
9 / 10	/ 20
Total	/ 100

This document consists of **26** printed pages, including this cover page and one blank page.

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Section A

Answer **all** the questions in this section.

- 1 Fig. 1.1 below shows 2 organelles from an animal cell involved in digestion. Assume that the cell surface membrane is located above the organelles.

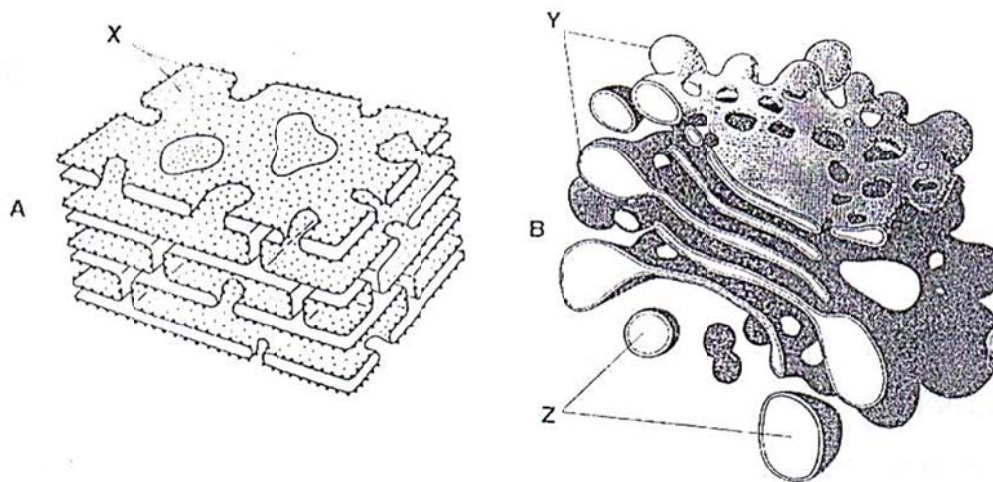


Fig. 1.1

- (a) With reference to Fig. 1.1,

(i) name structures **Y** and **Z** which are associated with organelle **B**.

Structure **Y**:

Structure **Z**: [1]

(ii) explain the roles and significance of **A** and **B** during protein synthesis .

.....

 [4]

(iii) describe how the contents of Y are subsequently secreted out of the cell.

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..... [2]

Fig. 1.2 below shows an electron micrograph of a eukaryotic cell.

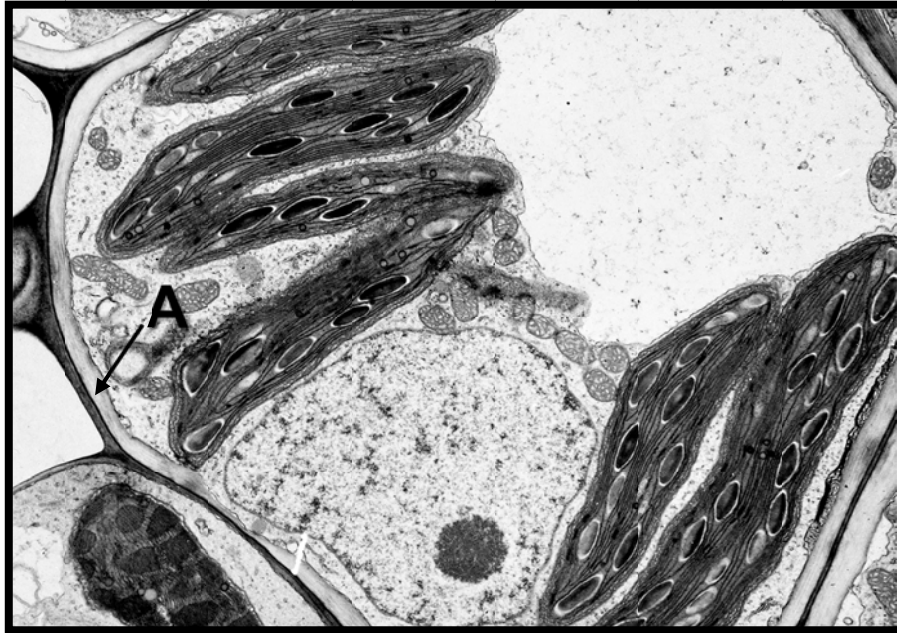


Fig 1.2

- (b)** (i) The primary structural component of A is known to be a polysaccharide. Name the monomer of this polysaccharide.

..... [1]

- (ii) In the space below, draw a labelled diagram to show how two such monomers are joined together.

[1]

- (c)** Collagen is the most abundant protein found in mammals. Unlike haemoglobin which serves a metabolic function, collagen serves a structural function.

Describe how the molecular structures of collagen and haemoglobin differ.

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.....

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[2]

[Total: 11m]

- 2 The graph below shows the DNA content during nuclear division.

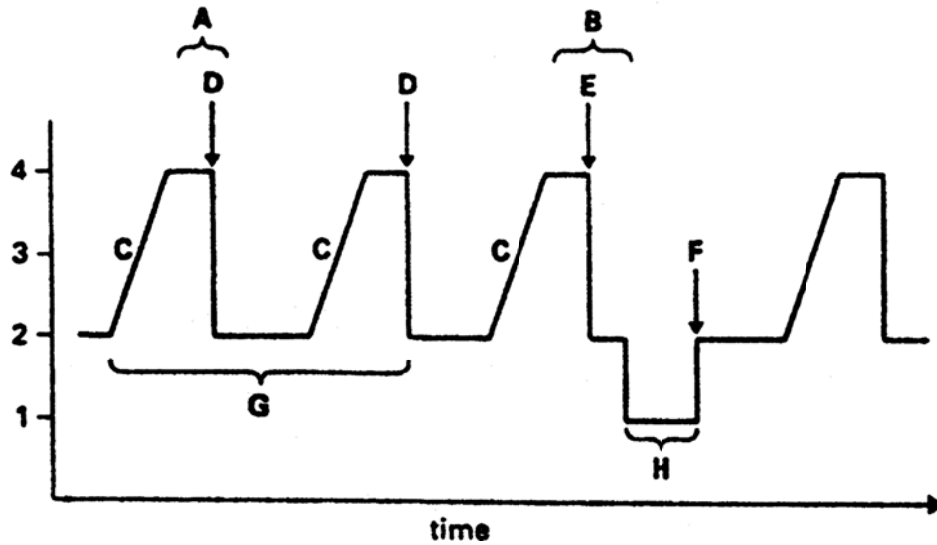


Fig. 2.1

- (a) State the stages D and E if y-axis of graph indicates DNA content per cell.

Stage D:

Stage E: [1]

- (b) Explain the importance of reduction division in meiosis to sexually reproducing organisms.

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..... [2]

Fig. 2.2 shows a linear chromosome undergoing the first round of DNA replication.

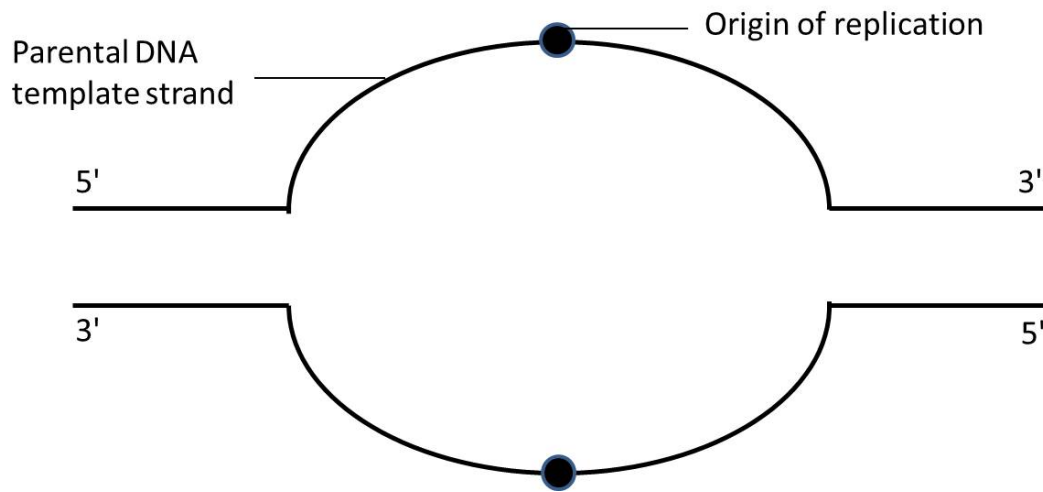


Fig. 2.2

- (c) (i) On Fig. 2.2, draw the direction of DNA synthesis for leading ($\rightarrow$) and lagging strand (---->).

[1]

- (ii) Describe two differences in the formation of the leading and lagging strands.

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.....

.....

[2]

In order to sequence the DNA of a gene, the DNA is first denatured to separate its two strands. Subsequently, in the presence of a large supply of each of the four nucleotides, the single-stranded DNA is replicated by DNA polymerase.

A low concentration of specially prepared nucleotides is also present. Once added to the chain, these nucleotides do **not** allow the chain to continue growing.

Each special nucleotide is labelled with a fluorescent dye, using a different colour for each of the four bases.

Fig. 2.3 shows a replicated DNA chain ending with one of the special nucleotides.

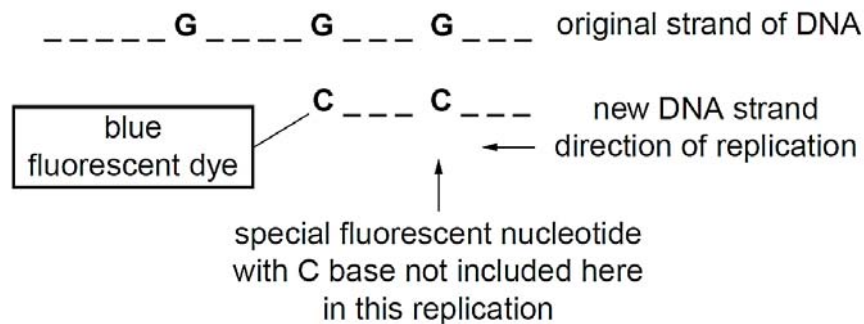


Fig. 2.3

- (d) With reference to Fig. 2.3 and to the information given, suggest why a special nucleotide with a C base was **not** included by DNA polymerase at the first site requiring a C nucleotide.

[1]

This method of sequencing a gene produces as many DNA fragments as there are nucleotides in the gene, each fragment differing in length by one nucleotide.

Fig. 2.4 shows part of a set of such fragments.

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      A_
     T__
    C___
   G____
  A_____
 T_____
  
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Fig. 2.4

These fragments are loaded onto a agarose gel, shown in Fig. 2.5, and separated by electrophoresis.

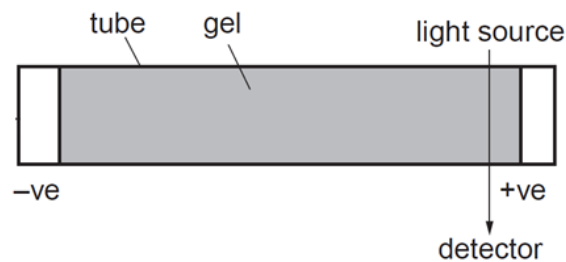


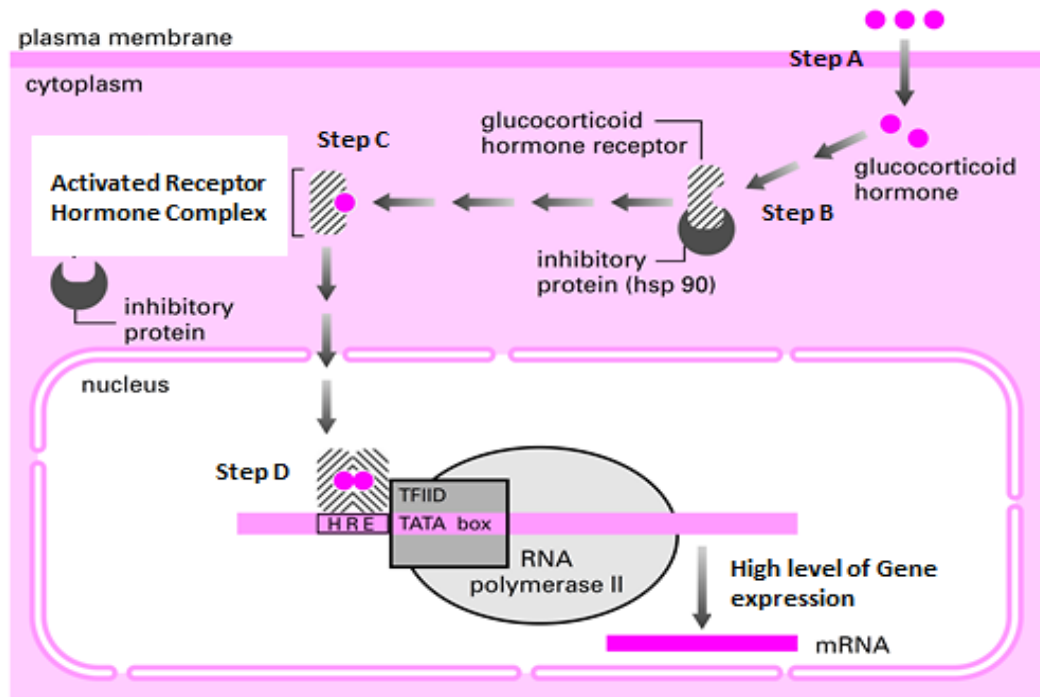
Fig. 2.5

- (e) In what order will the fragments shown in Fig. 2.4 reach the light source and detector shown in Fig. 2.5?

[1]

[Total: 8m]

- 3 Glucocorticoids are steroid hormones produced by the adrenal cortex that increase the transcription of several genes important in carbohydrate and protein metabolism. Fig. 3.1 below shows how Glucocorticoids can pass through the plasma membrane to enter the cytosol and bind to glucocorticoid hormone receptors. In the absence of glucocorticoid hormone, its receptor remains in the cytosol and is inactive.



Hormone response element (HRE)

Fig. 3.1

- (a) (i) Glucocorticoid hormone receptor is a class of transcription factors. With reference to Fig. 3.1, explain how receptor hormone complex (Step C) can be activated.

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..... [2]

- (ii) Suggest how Glucocorticoid hormone receptor is able to bind to the specific region of the DNA known as HRE site shown in step D of Fig. 3.1.

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..... [2]

(iii) With reference to Fig. 3.1, explain how high level of gene expression may be regulated by activated Receptor Hormone complex within the nucleus.

[3]

Fig. 3.2 below shows the various steps involved in the processing of primary RNA transcript.

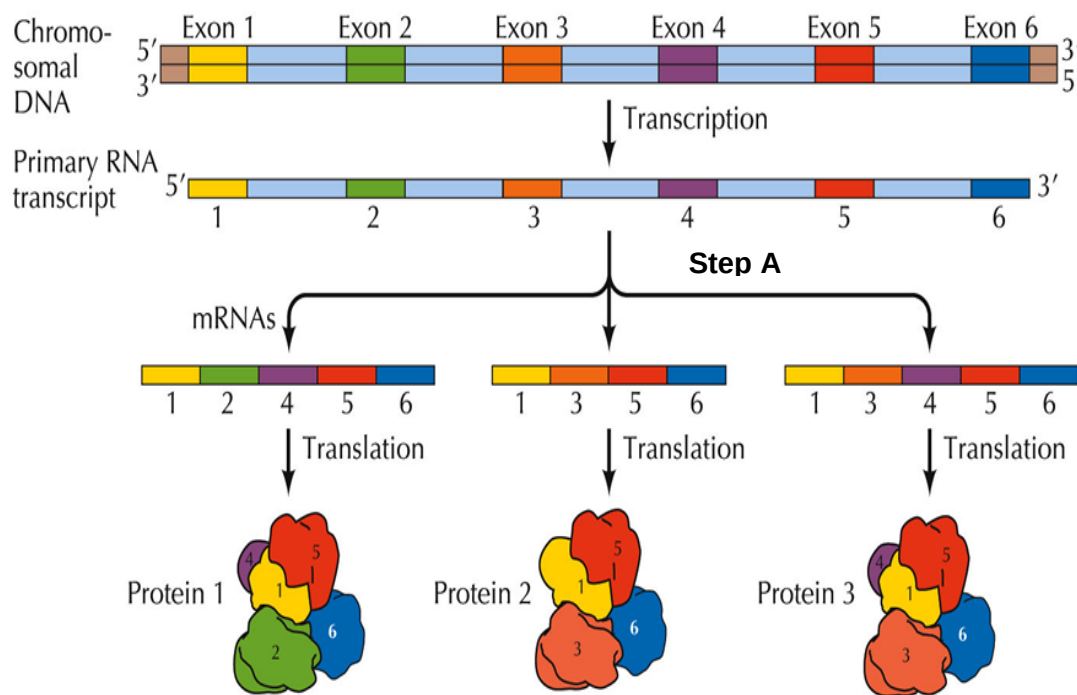


Fig. 3.2

(b) (i) Name process A and explain why it is essential in eukaryotic cells from an evolutionary view point.

[3]

(ii) Prior to completion of step A, the primary RNA is modified in several important ways. Explain how **one** of these ways helps to regulate gene expression.

[1]

(iii) With reference to Fig. 3.3, label the following region (label only once):

(i) 5' UTR

(ii) 3' UTR

(iii) coding sequence

(iv) promoter

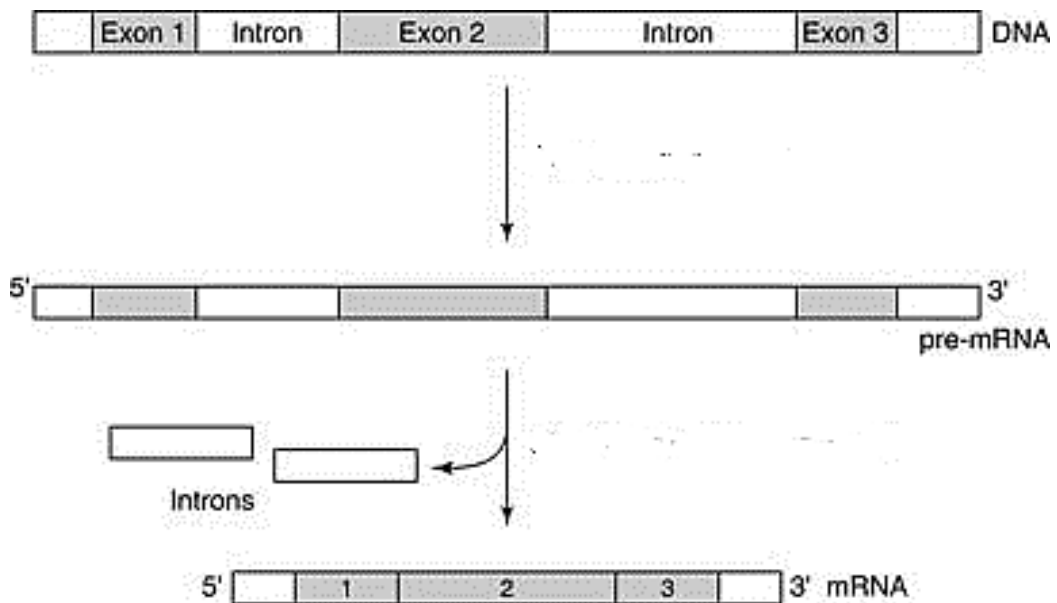


Fig. 3.3

[1]

[Total: 12 m]

- 4 Serotonin is a neurotransmitter which is produced by certain neurons in the brain. One of its effects is to increase the activity of sensory neurons in the brain. It also usually improves a person's mood and keeps the person awake. Fig. 4.1 below shows a synapse at which serotonin is the neurotransmitter.

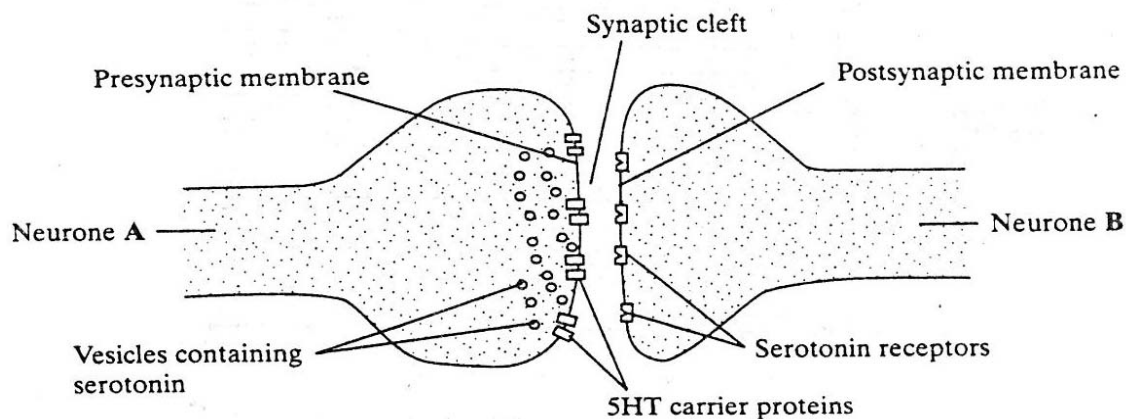


Fig. 4.1

- (a) (i) Explain how the release of the neurotransmitter, serotonin, results in the transmission of an impulse from neuron A to neuron B.

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[4]

- (ii) Serotonin is normally rapidly reabsorbed from the synaptic cleft by 5HT carrier proteins in the presynaptic membrane.

Suggest an advantage of rapidly reabsorbing the serotonin.

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[1]

The pain associated with osteoarthritis can be relieved using transcutaneous electrical nerve stimulation (TENS). It uses electrical impulses to stimulate the nerve endings at, or near, the site of the pain. Self-adhesive electrodes are stuck on the skin and attached to a small, portable power unit.

Fig. 4.2 shows a TENS machine in use.



Fig. 4.2

It is thought that TENS triggers the release of natural painkillers called endorphins, which are similar in shape to painkilling drugs such as morphine.

Fig. 4.3 shows synapses in a pain pathway from a damaged joint to the brain.

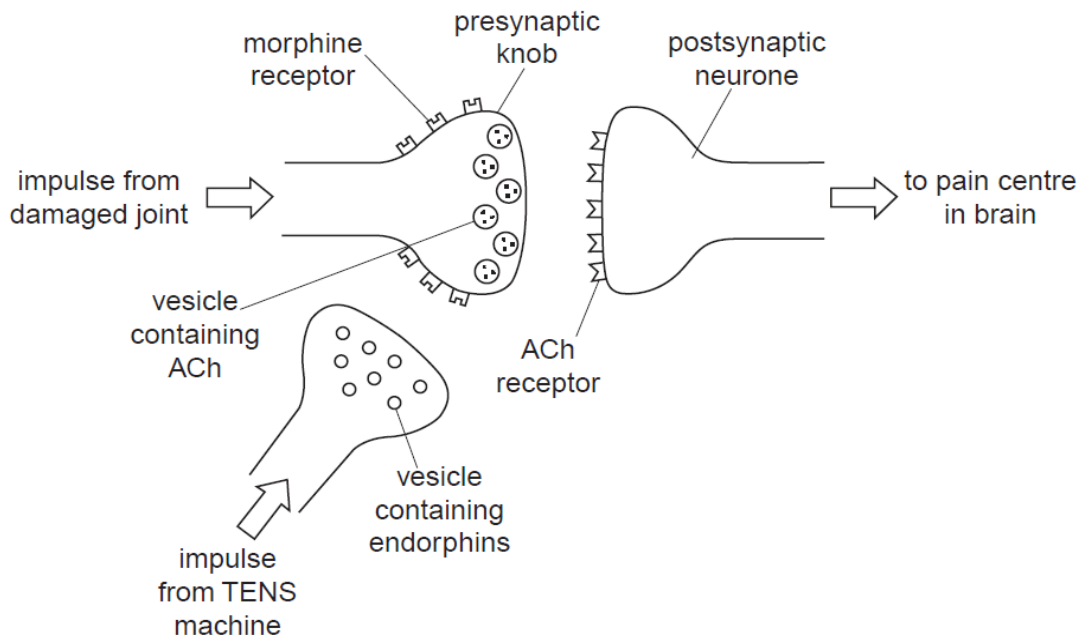


Fig. 4.3

- (b)** (i) Suggest how endorphins may act to reduce pain.

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..... [3]

- (c)** Some synapses in the brain use the neurotransmitter dopamine. After the postsynaptic membrane has been depolarised, dopamine leaves the receptor proteins and moves back into the presynaptic neurone through specific transporter proteins.

Schizophrenia is a condition in which there is a higher than usual concentration of dopamine in certain areas of the brain. The drug phenothiazine has a similar shape to dopamine and is used to treat schizophrenia.

Suggest and explain what occurs at the synapse when phenothiazine is used in the treatment of schizophrenia.

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..... [2]

- (c)** Explain the disadvantage of a signalling cascade

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..... [1]

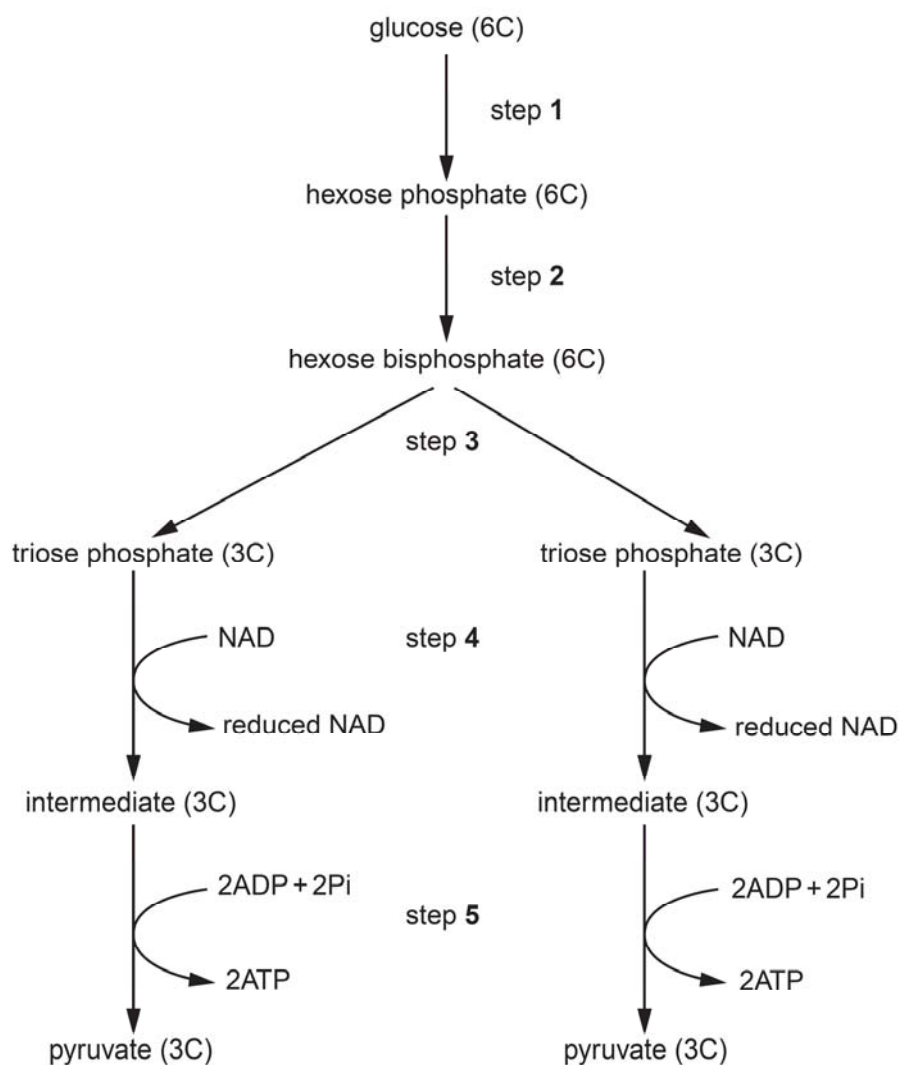
[Total: 11m]

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Section BAnswer **all** the questions in this section.

- 5 During the process of glycolysis, glucose is converted by a series of steps into two molecules of pyruvate. Fig. 5.1 outlines glycolysis.

**Fig. 5.1**

(a) With reference to Fig. 5.1, state the processes occurring at:

- (i) steps 1 and 2
- (ii) step 3
- (iii) step 4.

[2]

- (b) Pyruvate can enter a mitochondrion when oxygen is present. Describe what happens to pyruvate in a yeast cell when oxygen is **not** present.

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..... [2]

- (c) Fig. 5.2 outlines anaerobic respiration in yeast cells. Complete Fig. 5.2 by writing in the missing compounds.

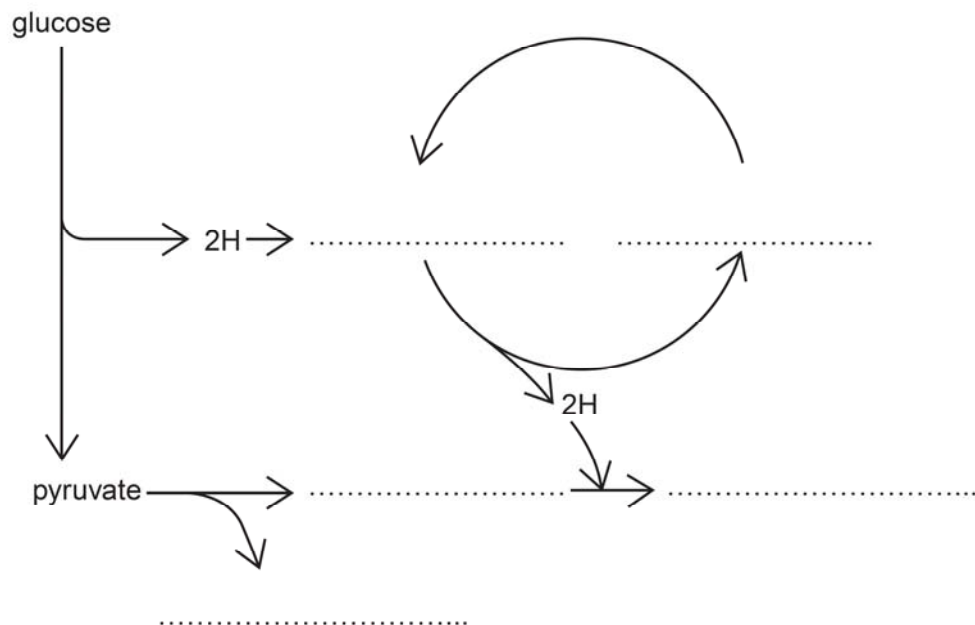


Fig. 5.2

[2]

- (d) Describe how anaerobic respiration in mammalian cells differs from anaerobic respiration in yeast cells.

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..... [2]

Fig. 5.3 shows the changes in the relative concentrations of RuBP and GP produced in the Calvin cycle before and after a light source is switched off. All other conditions are constant.

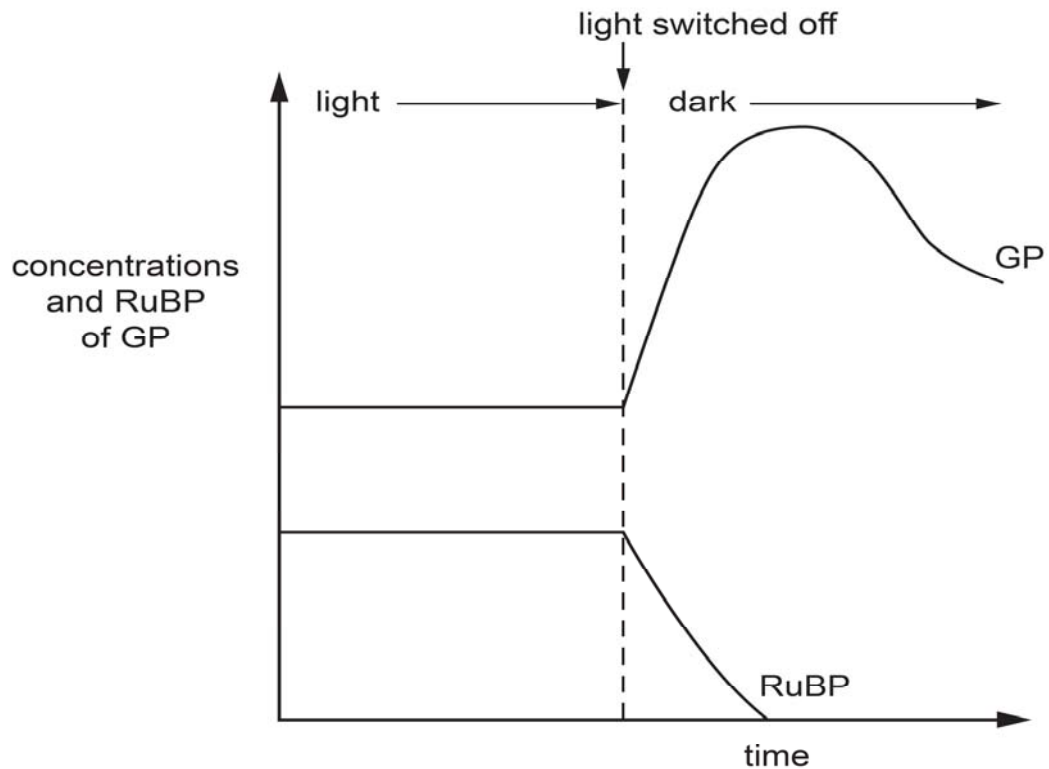


Fig. 5.3

- (e) Explain the changes in the relative concentrations of RuBP and GP **after** the light source is switched off.

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[3]

[Total: 11m]

- 6 It was observed that when a pure-breeding strain of squash that produces disc-shaped fruits was crossed with a pure-breeding strain having long fruits, all F₁ had disk-shaped fruits.

Once fully grown, the resulting F₁ plants were self-pollinated. The phenotypes of the fruits produced (shown in Fig. 6.1) were recorded in the following proportions:

Long	32
Sphere	178
Disc	270

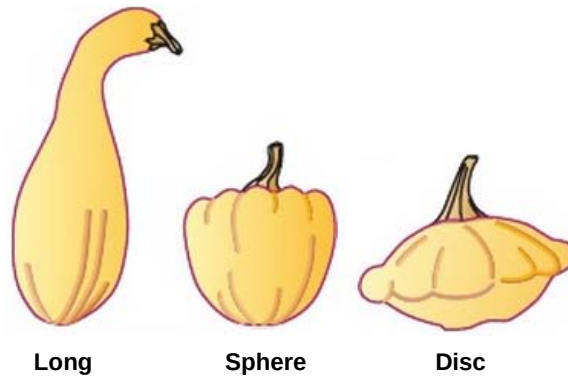


Fig. 6.1

- (a) Draw a genetic diagram in the space below to show the cross described and show the genotypes of the P, F₁, and F₂ generations.

[4]

(b) Explain how different genotypes give rise to different phenotypes in squash fruits.

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..... [2]

[Total: 6m]

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Section C

Answer **all** the questions in this section.

- 7 (a) In bacteria, it is common for two or more genes to be arranged together in an operon.

(i) Briefly discuss the arrangement of genetic sequences within an operon.

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..... [2]

(ii) State the biological advantage of an operon organization.

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..... [1]

- (b) The sequential use of two sugars by a bacterium is known as diauxic growth. It is a common phenomenon among many bacterial species. When glucose is one of the two sugars available, it is typical that the bacterium metabolises glucose first, and then a second sugar after the glucose has been used up. Among *E. coli* and related species, diauxic growth is regulated by intracellular cAMP levels and the catabolite activator protein (CAP).

(i) Summarise the effects of glucose and lactose on the ability of the lac repressor and the cAMP-CAP complex to regulate the lac operon.

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..... [3]

You have isolated a mutant strain of *E. coli* in which the *lac* operon is constitutively expressed. To understand the nature of this defect, you create a merozygote (diploid cell) in which the mutant strain contains an F' factor with a normal *lac* operon and a normal *lacI* gene. You then compare the mutant strain and the merozygote with regard to their β -galactosidase activities in the presence and absence of lactose.

You obtain the following results:

	Addition of lactose	Amount of B-galactosidase (percentage of mutant strain in the presence of lactose)
Mutant	No	100
Mutant	Yes	100
Merozygote	No	100
Merozygote	Yes	200

Fig. 7.1

(ii) Explain the nature and effect of the defect in the mutant strain.

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.....

[2]

- (c) Human Immunodeficiency Virus (HIV) is the cause of Acquired Immunodeficiency Syndrome (AIDS).

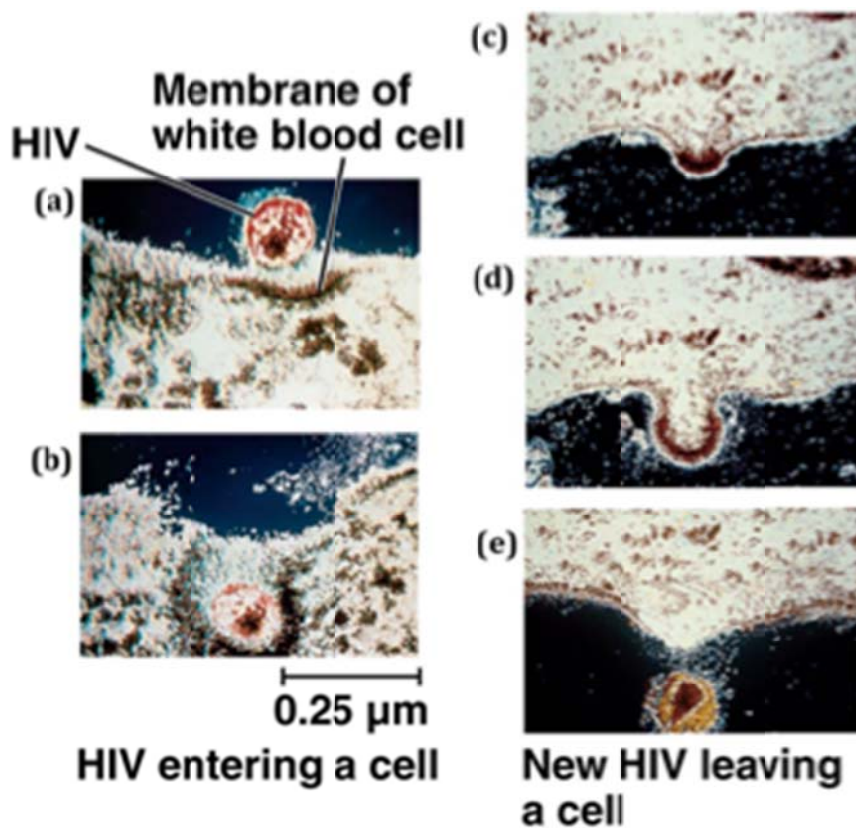


Fig. 7.2

- (i) Describe the events occurring between (a) and (b).

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..... [2]

- (ii) Although there are medications to prevent the replication of HIV, explain why there is no full cure.

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..... [1]

[Total: 11m]

- 8 Mammoths are extinct mammals related to elephants. About three million years ago, the ancestors of mammoths migrated from Africa into Europe and Asia. There, about 1.7 million years ago, the steppe mammoth (*Mammuthus trogontherii*) evolved and became adapted to the cooler conditions. Then, about 700 000 years ago, as the climate changed and the Arctic became much colder, the woolly mammoth (*Mammuthus primigenius*) evolved. Woolly mammoths showed a number of obvious adaptations to reduce heat loss, including thick fur, small ears and small tails.

- (a) Explain how variation and natural selection may have brought about the evolution of the woolly mammoth of small tail, small ears, and thick fur.

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[4]

- (b) A frozen, 43 000 year old woolly mammoth was found in Siberia. Its DNA was extracted and sequenced. The sequences of the genes coding for the α and β chains of haemoglobin were compared with those of modern Asian elephants. State two advantages of molecular methods in the study of evolutionary lineage.

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[2]

The map in Fig. 8.1 shows the mammoth migration and evolution around the world. Migratory events, either in the form of range expansion (natural movement away from parents) or jump dispersal (traveling long distance across inhospitable habitats), may lead to genetic isolation, thus facilitating speciation.

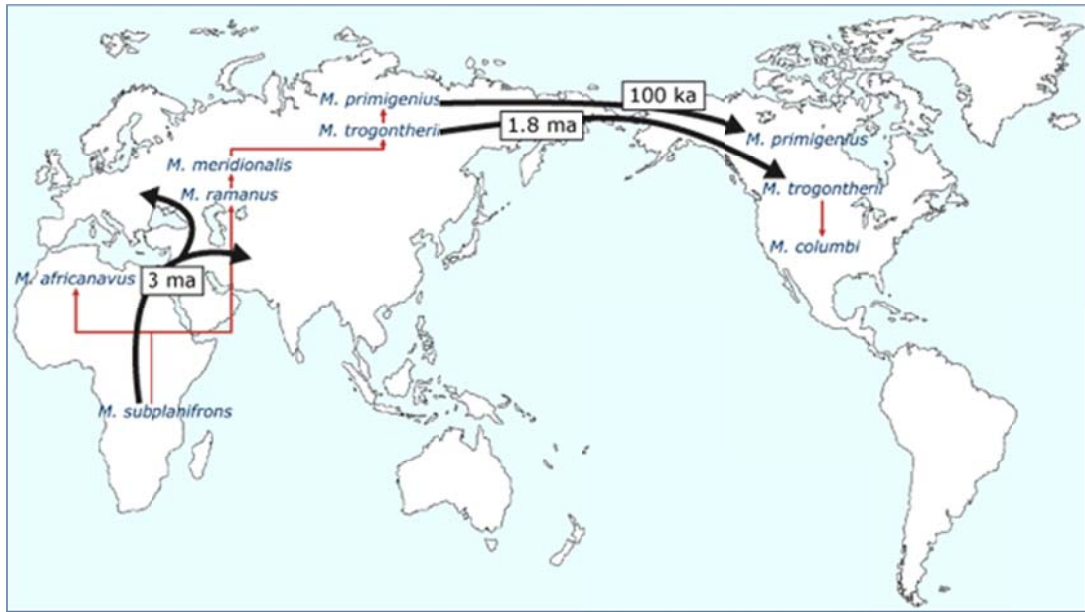


Fig. 8.1

- (c) (i) Suggest, with reason, which mode of speciation the mammoths underwent.

[1]

- (ii) State and describe two possible causes of sympatric speciation.

[2]

- (iii) Suggest why biological species concept cannot be used to classify mammoths into the different species.

[1]

[Total: 10m]

Section D

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

BEGIN EACH SECTION OF THE QUESTION ON A NEW PIECE OF ANSWER PAPER.

- 9** (a) Explain how fluidity of biological membranes can be maintained. [6]
- (b) Describe the main ways in which a globular protein differs from DNA. [8]
- (c) Explain, with examples, how anatomical and embryological homology supports Darwin's Theory of Natural Selection. [6]

[Total: 20]

- 10** (a) Describe the functions of **common proto-oncogenes** and **tumour suppressor genes** and explain **how loss of function mutation** and **gain of function mutation** can contribute to cancer. [6]
- (b) Explain how a change in the sequence of the DNA nucleotide may affect the amino acid sequence in a protein, and hence the phenotype of the organism. [8]
- (c) Describe how the structures of RNA are adapted for translation. [6]

[Total: 20]

---The End---